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DESIGN AS A CATALYST FOR THE CIRCULAR ECONOMY

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Introduction

Design *is* a powerful tool for change in the material world. We can design better products so that no materials go to waste, as in the natural economy, but this requires that the rationality, purpose, and direction to catalyse the circularity of economies is developed – not just in politics, but also in our educational programmes. This chapter's intention is to contribute insight into how design can be a catalyst for full circularity of products and materials. It presents both a review and critique of contemporary design's inability to address the complex problems of both today and the future that engage in closing the materials loop and going circular.

The chapter introduces a case about the design of typical solar cell panels and points out issues in relation to design for full circularity. The case is a real issue, a pedagogic case, which indicates the necessity to provide argumentation for change and show that by applying different sustainable design principles, new approaches can contribute to our thinking about design as a catalyst. This can be accomplished by specifically tying design strategies to how they might be used to effectively overcome the challenges of the current state of the take-make-dispose business models. One could argue that these design and production methods have grown out of a capitalistic economy that sees nature as a waste resource reservoir (Pearce & Turner, 1989). In a circular economy (CE), the focus is shifted from a treatment-of-waste strategy only to also include prevention-of-waste strategies. This is accomplished by applying certain principles, such as designing for disassembly to enable repairability, reuse, or recycling.

This chapter delves into available literature and draws upon the authors' experience in research-based teaching to identify and discuss directions for design as a catalyst for CEs. We look at the wicked problem posed by solar cell panels as an example of a product that generates renewable energy using green technology, that is not designed for end-of-life circularity. But we also discuss how they could be. Through design interventions, next-generation (photovoltaic) solar cell panels could be more sustainable at the materials, product, and service system levels. The case example is a typical design-challenging product that has provided the motivation (among many other cases) behind the development of a new design course, which aims to support a mindset for full circularity of products – one that will support an inner transition in educating and preparing designers for a sociotechnical transition towards a CE.

DOI: 10.4324/9781003267492-14
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The chapter discusses the implications, when teaching eco-design principles, of introducing new methodologies early in product development or when redesigning existing products. The design process is a complex cognitive process with several shared challenges. These emerge in the conflicts of routine versus disruption, old methodologies versus potential innovative ideas, and deductive problem-solving versus inductive idea generation.

The meaning of design and its potential as a catalyst for change

The word 'design' relates to the Latin word *designare*, which means 'to mark out' and has been used since early history to define how humans create, maintain, and transform the artificial, man-made environment (Cross, 2001; Moles & Jacobus, 1988). From the creation of simple artefacts and structures to intricate products, machines, and socio-technical systems, the history of design reflects on the co-evolution of knowledge, culture, social development, and the conditions of the environment with which we interact. So, it is not surprising that the meaning and use of the word 'design' is also in constant, context-dependent, re-definition, and diversification (Buchanan, 1992). Through the 18th and 19th centuries, industrialization deeply transformed both the artificial and the natural world and marked the beginning of design as a discipline, partly art and partly science, focusing on the acceleration of industrial, technological, and economic development (Buchanan, 1992). Several philosophical and theoretical movements followed this development, arguing either in favour or against the further alignment of design and the scientific method (Cross, 2001). In the early 1960s, the rise of increasingly complex problems, engendered by the cumulative effects of mass production and consumption, marked the further formalization of design as a *problem-solving* methodology widely applied in engineering and social sciences.

Design principles are more than creativity

Since the 1960s, there has been both a diversification of design activities in engineering, businesses, government, and other social services (Ceschin & Gaziulusoy, 2016; Manceau & Morand, 2014), and the reemergence of debates about the design-science relationship. As an example, Cross (2001, p. 5) claims that whereas scientific methods need to be repeatable, a method of design practice need not be. In most cases, it would be preferable that it cannot be copied. Considering this contrast to the epistemology of science, Cross (2001) argues that the theory of design has little to learn from science, and that it is better positioned to undertake the task of developing the *logic* of creativity. An argument against this perspective emerges when we perceive design as a means to achieve sustainability. Importantly, a design is made of more than creativity, and more than sustainability; it is about how to integrate the two, while ensuring functionality (i.e., mechanical engineering, quality, product standards), value properties, and aesthetics. We claim that no other discipline than design holds so much potential to excel in working with constraints.

This argument brings up another perspective on methodologies; Buchanan (1992) emphasises flexibility as one of the most important attributes of the design methodology when approaching problem-solving tasks. He says that in design, flexibility emerges from eluding reductionism (intrinsic to the scientific enquiry) and from the availability of various methodologies and principles that support the practice. Another concurrent debate pertains to the role of creativity in the design process. Dorst and Cross (2001) argue that, although it is very difficult to verify the occurrence of the creative event during the design process to assess the novelty of its outcomes, creativity is present in any design project, either as an explicit event (e.g., breakthrough) or as the

implicit value that unique solutions might acquire over time. Runco and Jaeger (2012) suggest that the problem with, or perhaps confusion about, creativity in design derives from the lack of a standard definition linking originality and effectiveness. They conclude that originality, the basic attribute of a creative act, is not enough to ensure that a novel product or solution is in fact useful or valuable, therefore, creativity in design must also either have or be guided by appropriateness. Such distinctions have acquired notoriety in recent decades, as more design principles and methods are radically influencing the outcomes of the processes of innovation in circular economies and strategic decision-making in both private and public organizations, (Ceschin & Gaziulusoy, 2016; Liedtka, 2015).

During the last decade, much attention has been given to an emerging problem-solving approach known as Design Thinking (DT). Liedtka (2015) has indicated that although DT was originally used to improve the quality of the product development process and its outcomes, it is currently applied in the development of services, strategies, policies, education, and other social services. In practice, DT is a transposition of the creative way designers solve problems into a hypothesis-driven process of "how-might-we" questions that focus on the dual nature of human thinking, dynamically assessing the problem, multiple alternative solutions, and the tensions between constraints and possibilities through abduction and experimentation (Liedtka, 2015).

DT is particularly useful when decisions are made under conditions of high uncertainty or ambiguity, and when active iteration through learning and/or experimentation is required. However, we argue that the traditional design approach, serving foremost as a process for creativity, is not enough to guide system transformation; it needs methodologies and valued principles, not just at a product level, but from a product-service and system perspective. There have been several initiatives at the macro-level to push for more sustainability in our consumption and in our production systems. The United Nation's Sustainable Development Goal (SDG) no. 12, entitled "Responsible consumption and production", focuses on changing unsustainable patterns of consumption and production to achieve global sustainable development, and all major groups should play an active role. (United Nations, 2021). This goal emphasises the need for national and regional initiatives to accelerate a shift to promote social, environmental, and economic development.

Our recommendations on how design can be a catalyst for a sustainable CE are approached from various perspectives, for example, through the evolutionary path of eco-design practice and theory, and critique of previous and present design-thinking approaches when the primary focus is on the logic of creativity alone and not on integration with the logic of sustainability.

The logic of sustainability is approached from the perspective of design for CE and an example is illustrated by solar cell panels (photovoltaics). A solar cell panel represents a critical and illustrative example of not only the need to understand innovation opportunity at the product level, but also of the need for alignment at the policy level, meaning how policy frameworks can push for innovation and products that are better prepared for a CE than those we currently have.

Designing for a circular economy

In recent decades, one significant example of the co-evolution of design and developmental history has been the emergence of dedicated design principles, techniques, and methodologies for addressing the contemporary challenge of sustainable development. This section provides a brief discussion on the evolution of two concepts that are particularly relevant to the contemporary role of design in the context of socio-technical transitions: *sustainability* and *circularity.*

Design for sustainability

The concept of design for sustainability, although vague and with more than a hundred different definitions (White, 2013), has been roughly operationalised by integrating environmental requirements early in the design process of products, along with common social and economic needs. Initially, sustainability in design focused on reducing potential environmental impacts and the environmental footprint – the intensity of consumption of natural resources and services of a product over all stages of its life cycle, from important decisions taken in the design phase, during its production, and in its use to the end of life. This life cycle approach gives the designer an important responsibility when considering, visualizing, and selecting materials and manufacturing methods to design for a specified lifetime.

Sustainability in design has been all about minimizing the negative impact on the environment through eco-efficiency However, the critique is that eco-efficiency takes the linear approach to make the cradle-to-grave models more efficient by seeking methods to minimise the energy, volume, velocity, and toxicity of the material flow system (Braungart et al., 2007). Yes, materials can be recycled, but often as an end-of-pipe solution, because the materials were not selected nor the mix of materials designed for recycling, and therefore it most often ends as a downcycling, such as a degrading of the material quality by crushing or incinerating. Every year, we overshoot planet Earth's regenerated biological resources earlier than the year before. In this paradigm, the slopes will not be so steep on the Earth's minerals and metals, but the sources of the problems will not be affected. The benefit of the introduction of eco-efficiency is that it will make a clear difference in the short run, but in the long run, it will not change much.

Eco-design principles

The EU Commission introduced the Ecodesign Directive in 2009 to reduce the consumption of virgin materials and natural resources by decreasing energy consumption and increasing the recyclability of products. The Ecodesign Directive focused primarily on energy consumption, use of toxic substances, and the end-of-life (EOL) perspective on energy-related products (ERPs). Since 2015, materials are an integrated part of CE ideas that favour requirements for material efficiency, thus overcoming the previous critique of design for sustainability alone being an insufficient approach. In the M/543 Mandate of the EU Commission, the delay in the inclusion of material efficiency is partly explained by the absence of adequate metrics. "These include absence of standards for assessing material efficiency aspects identified in previous product specific Ecodesign implementing measures" (EU Commission, 2015, p. 2). According to the mandate: "An increased focus on material and resource efficiency aspects in the application of Directive 2009/125/EC should make a sizeable contribution to the transition towards a more CE by making consumer goods more durable, resource-efficient and recyclable" (EU Commission, 2015, p. 3).

The standardization request on material efficiency in the Ecodesign Directive is linked to three aspects of value supply in designing for sustainability: extending product lifetime, the ability to reuse components or recycle materials from products at their EOL, and the use of reused components and/or recycled materials in products. The M/543 Mandate presents a driver in the Ecodesign Directive for designing with a material-efficient approach in product development. The extension of the Ecodesign Directive and the material efficiency mandate to more product categories provides a framework that will allow producers to consider the environmental impact in a product's life cycle from the initial design stage to mitigate the negative influence of the product on the environment. One example is the recent inclusion of electronics, such as smartphones

and tablets, with energy-efficiency transparency on the battery and the consumer's right to repair and change the battery if the battery is only chargeable up to 500 cycles. Whereas, if the charging function is possible for 1,000 cycles, then the product's battery can be designed and specified for professional replacement.

Although the Ecodesign Directive is complementary in use and purpose, due to the absence of an overarching design philosophy in sustainability to unify the design task, organizations will tend to use these tools indistinctly, to adapt the design process to external changes in national or international policy and regulations (den Hollander et al., 2017), or to shift consumption preferences guided by cultural and social purchase motives. As a result, the design tends to focus on a reactive optimization of the environmental performance of a product rather than on proactive innovation.

Design for cradle-to-cradle

The cradle-to-cradle perspective is a current global product standard, where product owners can achieve a certificate on their products. It is a private certification, yet highly a recognised one. It is a standardised type of 'product due diligence' across five categories of sustainability performance: (1) the materials used are safe for people and the environment, (2) the design enables regenerative product and process design for circularity, (3) the product design is protecting clean air, promoting renewable energy, and reducing harmful emissions, (4) the product is designed, produced, and can be circulated while safeguarding clean water and healthy soils, and finally, (5) the product's life cycle upholds social fairness by respecting human rights.

In contrast to eco-efficiency in the sustainability framework, the concept of cradle-to-cradle introduces eco-effective designs that strive to create material flows that create a workable relationship between the ecological systems and economic growth. Instead of a cradle-to-grave flow of materials, eco-effective design creates flows in which materials are used repeatedly at a high level of quality, and thus forming a cradle-to-cradle perspective. Eco-effective design is an integrated part of CE thinking as an approach that will take time to establish, since it requires design changes at the system level. Cradle-to-cradle design includes the creation of new value chains, business models, and supply chains that will have to be established. In the long run, the aim is that such an economy will allow the recirculation and reuse of materials, thus changing the future impact to establish the full CE (Braungart et al., 2007).

Design for full circularity

The circularity of products and materials is framed within a large and complex challenge, shifting from a linear economy to a CE. This shift implies looking at circularity not as one more economic function, but as a new economic paradigm. This means that we need to solve the challenge of economic growth, so that it becomes decoupled from the fast consumption of natural resources, and the vast amount of waste that goes to landfills or incineration is eliminated or at least slowed down by maintaining products, components, and materials at their highest utilities and values (Dokter et al., 2021). For design practice, this implies the further diversification of the Design for X (DfX) toolbox of Design for Durability, Serviceability, and Disassembly, and the adoption of new guiding principles for technological upgradability and energy efficiency.

The foundations for CE have emerged from the synthesis of several schools of thought. Major contributors include Walter Stahel, who introduced the service, or performance, economy

proposed in his *Inertia Principle*: "Do not repair what is not broken, do not remanufacture something that can be repaired, do not recycle a product that can be remanufactured. Replace or treat only the smallest possible part to maintain the existing economic value" (Stahel, 2010, p. 195). Other giants who have played a central role in laying the foundation for schools of thought are William McDonough and Michael Braungart with their cradle-to-cradle design philosophy, Reid Lifset and Thomas Graedel with industrial ecology, and Amory and Hunter Lovins's natural capitalism. But it was Gunter Pauli (2011), with the blue economy, who noted that the world market for waste and recycling was worth USD$450,000 million in 2010, while Nam (2015) estimated that the market for "reusing those materials takes up 0.01% of the overall market" (Ahn & Lee, 2018, p. 3). However, circularity is not without challenges. Bocken et al. (2016) argue that establishing CE flows requires that recycled materials have properties equivalent to the original ones; they claim this can only be accomplished through de/re-polymerization or closed-loop recycling. This is a complicated task and can be considered as a utopian approach, because all materials will degrade over time.

To catalyse a product's material circularity calls for the consideration of not only one product purpose during the design phase but in designing the product for a second or even a third commercial afterlife. For den Hollander et al. (2017), the operationalization of this principle in design refers to maintaining a product's *integrity* – its original form and function at manufacturing – if possible, in circular use, which would reduce, or even eliminate, the environmental cost of any intervention needed to maintain or restore the product's added economic value. However, shifts in cultural or socio-technical conditions can render the use, appeal, or value of a durable product as obsolete, limiting value retention options to recycling and upcycling. In this case, upcycling – the use of discarded materials to create a new object of higher value (Sung, 2015) – is preferred over recycling. Nevertheless, as an economic activity, upcycling is not yet as well-established as recycling.

In general, complex durable products are often easy to repair or refurbish but much more difficult to recycle (den Hollander et al., 2017). Still, even if discarded products, components, or materials can be fully recycled, value retention might still be limited by cultural, economic, or technical barriers to the use of recycled materials, recyclates, in manufacturing new products. Although unused recyclates can be stored until further use becomes possible, in large quantities and over undetermined periods of time, this is no different than hoarding. If materials do not circulate, new material inputs will be required to supply demand. To survive as an economic service, the recycling activity needs to achieve and sustain a considerable and constant flow of materials and recyclates. However, this is in opposition to the goal of increasing product value retention through durability and integrity because recycling is basically a destructive activity (den Hollander et al., 2017). These important conceptual contradictions emerge from the approach of the current DfX toolbox, since each technique focuses on one aspect or activity in the CE independently, and not on circularity itself; this leaves many choices open for the designer.

The subsequent introduction and popularization of the concept of circularity in business organizations encourages focusing on the optimization of quality products and towards innovation of products and services. In its most basic form, circularity in businesses means keeping the flow of products and materials circulating, and if possible, to increase value retention by mimicking the efficiency of ecological cycles during which nothing is wasted. Value retention in products is commonly achieved through design that prolongs the product usage stage through reuse, repair, refurbishment, or remanufacturing, while value retention in materials commonly refers to recycling at the product's EOL (Ceschin & Gaziulusoy, 2016; den Hollander et al., 2017; Dokter et al.,

2021). In this context, the role of design focuses not only on product development, but also on the development of new services, function, and processes. However, den Hollander et al. (2017) argue that in practice, circularity often focuses on material retention, recycling. This is a less costly and challenging type of innovation from the perspective of both design and management, but also a far less valuable option than product retention in terms of environmental and economic sustainability. In this context, one would argue that if circularity is based only on material retention, it is not sustainable. A product that cannot be recycled for reuse in some form at its EOL does not hold value.

When designing new products or redesigning exiting ones, we must remember that design principles and approaches are currently the only means to solve design problems, and the extent and value of design solutions are only as great as the scope of the problem they attempt to solve. Ceschin and Gaziulusoy (2016, p. 119) suggest that sustainability is currently understood as "a system property and not a property of individual elements of systems" that involve a process-based, multi-scale and systemic approach to planning, rather than the usual goal-based optimization. This distinction is relevant. For example, when we discuss a CE, we also focus on nature's resources for a sustainable future; when we talk about green conversion, climate, and nature's resources, we must make sustainable designs and avoid toxic materials, use fewer materials, and make sure we can separate the materials afterwards in clean fractions (i.e., design for disassembly), so that they can be recovered and recycled, thus avoiding landfills and incineration.

In the next section, we show how a design for circularity can be used to assess products and thereby innovation opportunities at the product level and the manufacturing level. We also show how policy frameworks, if they are aligned, can be a catalyst for change in design – provided, of course, that the consumer or industry is willing to pay a little more for the most sustainable solution for the future or at least during the transition period.

Example: Solar cell panels for circular design

Solar power is one of the champions in the green transitions that are shifting from coal, oil, and gas. But who really knows what happens to solar cell panels when they no longer work efficiently and need to be scrapped? The solar cell panel is a typical and challenging example of transitioning towards a CE.

The International Renewable Energy Agency estimates that by 2050, there will be 78 million tons of solar-panel waste in the world. Today, 85% of the material used in solar cells can be recycled (Heath et al., 2020); the remaining material is scrap. From a design-for-recycling perspective, the issue is that some of these products' materials are glued or melted together making some part unrecyclable or even not allowed to be incinerated as the process would release unhealthy chemicals (SDU Press and News, 2021). Figure 12.1 show a 3D model of a typical solar cell panel illustrated using Autodesk Inventor (2022). In the following, we will refer to 'the backside' foil and 'the stringed solar cell', when discussing issues such as the material composition of the backside foil and lead in the soldering connecting cells.

Issues with the current design

Two design issues that block a circularity strategy can be mentioned, because even though we have the knowledge that certain chemical compounds are unhealthy, they are used anyway because they are legal, cheap, and very functional.

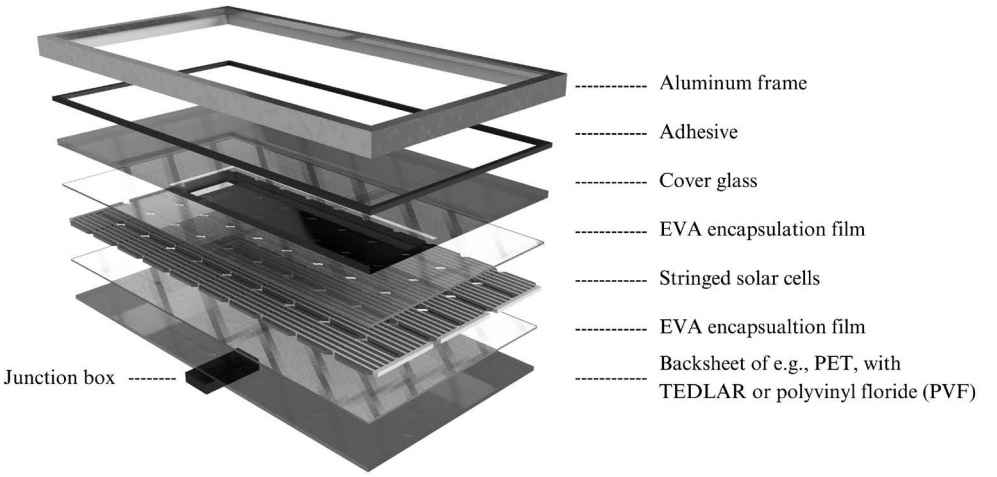


Figure 12.1 The sandwich layers of a solar cell panel, 3D model.

Issue one is the backside foil that is designed to last but not for recycling. Some of the first-generation solar cell panels consisted of glass panels on both the front and back sides of the unit, but to make them more lightweight, most manufacturers have substituted the backside glass panel with a foil – either a polyethylene terephthalate (PET) polymer (what water bottles are made of) or a polyvinyl fluoride (PVF) polymer. To compensate for the glass protective side for the electronics, most foils are made of PVF – a material with excellent protective functional properties for the outside. But the backside foil is glued together to the stringed solar cells by the tough EVA (ethylene-vinyl acetate) encapsulation film, which is making it hard to separate these parts as to prepare for recycling (Fiandra et. al, 2019). Materials like the backside foil cannot be easily recycled (often too costly) and incineration, is only an option if the incinerator is fitted to scrub out hydrogen fluoride. So, the backside foils are currently placed in landfill deposits as the "best" option for their disposal.

Issue two is the use of lead in the soldering. Lead is a toxic heavy metal that accumulates in nature and in humans; the EU has banned lead in nearly all electronic products. Most photovoltaic (PV) products or solar cell panels contain lead, as these products are still excused from the EU lead-free directive. According to *PV Magazine*, which is a leading international solar cell magazine, a typical 60-cell crystalline silicon solar module contains up to 12 grams of lead (Hutchins, 2019). As long as the lead is in the solar cell panels, it is not dangerous for the environment. However, factory workers are exposed to it during manufacturing, and if solar cells are not handled correctly in recycling, there is a risk that the heavy metal will be released as dust particles. It can be argued that the exemption in EU legislation is from the time when solar energy had to compete with fossil fuels in terms of price and efficiency, and lead was the solution to make solar cell panels cheap and energy effective. But today, the situation is different. It seems unnecessary to use lead in the solar panel soldering. The electronics and automobile industries have shown us that it is possible to manufacture complex products without lead. Retaining this exemption for solar cell panels does not exactly drive the innovation and development of new solutions. As Hutchins (2019) argues, there are several alternatives that could completely replace lead, but the problem is that PV manufacturers are not motivated to replace lead with more expensive solutions, such as tin, when it is still legal to use lead.

Policy support to design out unnecessary chemicals

It is obvious that EU policymakers should reconsider the necessity to continue to exempt solar cell panels from the general legislation applying to electronics, thus pushing the solar cell panel manufacturers to compliance.

The International Technology Roadmap for Photovoltaic (ITRPV, 2022) publishes a yearly report overseeing the market share for solar cell technologies. In this report there is among others an overseeing of also the market share of different cell-connecting technologies. As the diagram of the ITRPV 2022 forecast in Figure 12.2 shows, solders that contain lead are today the standard technology. Even though lead is being phased out by producers, lead is still expected to be present in over 55% of the solar cell contacts in 2030. In the ITRPV 2019 forecast it was expected to be 50% by 2030. It is good news that phasing out is underway, but this transition moves slowly in relation to how fast we want the green transition to move.

How design for circular economies can fix the problem

To be sustainable, products also need to be designed with eco-friendly design principles, such as designing out chemicals that may present a barrier at the EOL to either reuse or recycling. We could say that the problems design has created, are also problems that sustainable design can fix.

For example, using material from scrapped solar cell panels to make new ones is still problematic, so how can we make reuse easier in the future? For one thing, we could design solar cell panels that avoid using toxic materials such as lead. We can use fewer materials as to design for durability and not mortality, and we can make sure materials can be separated afterwards so they can be reused at a higher value than now (i.e., reuse of silicon in new panels) instead of being crushed and downcycled. Particularly, the EVA glue film (ethylene-vinyl acetate) that binds the solar cell together and keeps it robust in all kinds of weather makes is incredibly difficult to disassemble the panel in large elements.

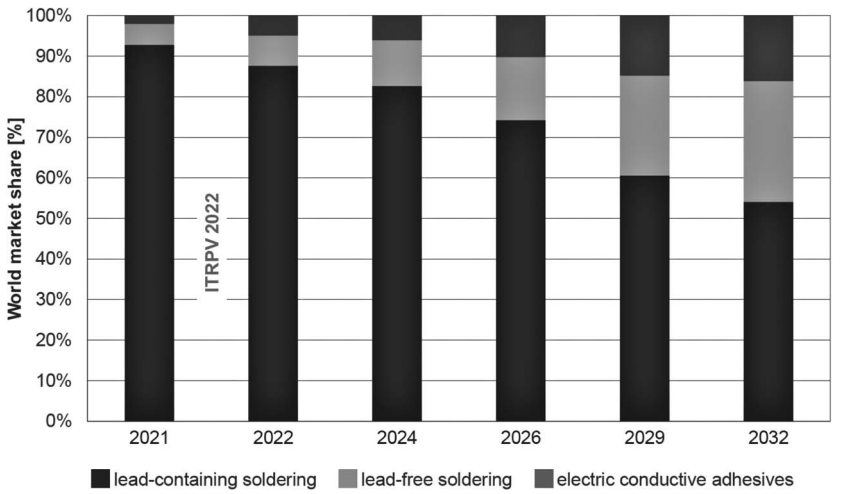


Figure 12.2 Expected market share for different cell-connecting technologies, ITRPV 2022 forecast.

Today, wires are removed, and aluminium from the frames recycled, while some of the remaining material is crushed and used in the steel industry or as road fill. Much of the glass is crushed into small fractions, leaving mixed residues consisting of common polymers, non-polymers, and residues of lead, silver, and copper, which have not been separated before crushing. These mixed fractions are not very valuable; they cannot be recycled for use in new solar cells, because it is cheaper for manufacturers to buy new materials. If a way could be found to dissolve the incredibly strong polymer lamination, the glue between glass and solar cells, it would then be possible to recycle more silicon and glass in whole and separate parts. This could create a market at a higher value for reusing more of the fully moulded materials. Recycling these materials would utilise the resources of the waste material, but also increase the product's carbon footprint. The argument, however, is that recycling is not likely to emit more carbon than the current process of both acquiring new raw materials and processing them. Repurposing the product is the alternative for retired material, but products are generally not designed for that either.

The solar cell panel is a very typical example and a strong motivator for the need to learn from the problems that we face from the linear design thinking approach, and then make a change in educating for the future when designing for circularity. It is no longer enough to point to the wider ecosystems and socioeconomic developments to assess and address their current problems. There is also an urgent need for capacity building. We have understood that the teaching of sustainability in universities needs more integral approaches, and this motivated us to develop a new intensive course for our design engineers named 'Sustainable Materials in Product Creation'. In the next section, we share the background motivation and the course's learning objectives.

Education: Designing the inner transformation

Universities play a vital role in capacity building because education is a tool for inner transformation. In the publication "*Engineer of 2020*" (US National Academy of Engineering, 2004), problems related to climate change and the environment were expected to become critical in 2020, and solutions to sustainability were expected to be a high priority focus for engineers' aptitude and ingenuity. Engineers were also anticipated to have a strong ability to learn new things and to apply their knowledge to entirely new problems and contexts, and at a faster pace than earlier generations. Although academic preparedness for attention to environmental developments was proclaimed a high priority a decade earlier, a study from Chalmers University in Sweden supported the prospect that companies would experience a lack of knowledge in sustainable development among newly graduated engineers (Hanning et al., 2012). Already then, scholars pointed to more engineering courses integrating sustainability as part of the solution. Guerra (2017) argued that a successful integration of sustainable development in engineering curriculums would require an approach in which students develop a holistic and systemic view of technical problem-solving that uses sustainability principles and applies a broad framework that covers both environmental, economic, and social contexts. However, each of the various types of engineers in an industrial organisation often works within its own specialised issues (Gericke et al., 2020). This leads to what is known as working in silos, which is described as one of the key challenges for improving the absorption of new methods. Design methods often aim directly at creativity, while the indirect link between systematic design methods and innovation is often disregarded. Junyent and Geli de Ciurana (2008) advocate a shift in the engineering educational paradigm by highlighting the importance of contextualization in teaching. With an ambition to break down silos and integrate CE with Design Thinking, we created a new design course for engineers.

The experience of creating a design course for engineers in a circular economy

During the spring semester of 2020, the course 'Sustainable Materials in Product Creation' (SMPC) was implemented for the first time as a part of the master's level engineering programme in product development and innovation (PDI) at the University of Southern Denmark. The ambition behind the new course was to educate engineers to think about sustainability from the outset.

The main idea behind starting the course is that designers and product developers play a key part in developing more sustainable products. As the EU Ecodesign Directive has recently pointed out, up to 80% of a product life cycle's sustainability impact is determined at the design stage. In the course, students work with thinking in circular value chains and with how to develop products with CE principles in mind. Thus, the students work with designing the entire product life cycle and not just the product itself. This means that the students need to have an extensive knowledge of the materials used in the products they develop.

The course integrates eco-design and green engineering principles in the didactics of problem-based learning. The learning objective of this approach was to provide students with insight into the different aspects of sustainable design and an understanding of limitations and opportunities shaped by materials when they considered sustainability and the possibility of circular material flows. The aim was also to engage the students with companies in several innovative processes, where they focused on re-creating or redesigning products and/or creating new products from upcycled materials. For the learning objectives, see Figure 12.3.

The intention in rethinking this master's course was to prepare the candidates for a future where they think beyond the traditional uses of materials. Instead, they should think in terms

This course will assist students to develop their understanding and knowledge of:
○ Recycled and sustainable materials in the product creation process. ○ The practical considerations of incorporating sustainable materials into the production process. ○ Methods of reflective practice in the field of product design.
Students will have multiple opportunities to apply and reflect upon their ability to:
○ Explore approaches to understanding material selection by gaining real life input from external companies. ○ Plan creative product design concepts and prototypes as a strategic approach to understand the limitations and opportunities of sustainable materials. ○ Effectively communicate an understanding of value added via sustainable materials.
In this course, students will work in teams to challenge themselves and so develop competencies in:
○ Practical project management. ○ Identifying and initiating action on opportunities that emerge from real world research and practical experimentation. ○ Reflecting on the role of sustainable materials in the entire product lifecycle. ○ The ability to feed test data into the product value creation process.

Figure 12.3 Learning objectives: Master of science course in sustainable materials in product creation.

of circular material flow and develop new methods to make the best possible use of materials. The course involves group work, participatory teaching, and curiosity-driven research. Through the company cases that the students work on, the aim is that they can apply the knowledge they acquire from the course to real-world sustainability challenges. Through their work with sustainable materials and design of the entire product life cycle, students become better equipped to handle the world of change they encounter when they enter the job market – precisely because so many companies, either right now or in the future, face challenges when it comes to restructuring to the use of more sustainable production methods.

In the course, students are split into groups, in which they participate in an array of workshops. Through these workshops, the students learn about the repurposing of materials, system thinking, and CE. When developing the course, we contacted companies with expertise in sustainable development and the reuse of materials. In their groups, the students work with concrete cases through cooperation with real businesses. To solve these cases, the students must come up with ways to repurpose materials from the businesses' products. Through the cases, students gain an extensive knowledge of materials, reflect on the challenges of recycling, and gain experience with building concepts and mock-ups for visualising their ideas.

By letting the students work in groups with company research teams, the aim of the course was to let the students experience what it is like to work in both a practical and academic way with material driven product development. Furthermore, it gave the students a way to critically reflect on how products can be designed in a more environmentally friendly way even today, and how the materials we choose in the design phase have an impact on whether materials can be reused or whether they just end up in a landfill.

Material design thinking for circularity

A Material Driven Design approach and system thinking are introduced as a guide to design based on waste materials or to investigate the CO_2 bill of materials for a given product and the materials' recyclability. This method contradicts their normal way of thinking, which is usually linked to the three phases of the DT method: 'inspiration', 'ideation', and 'action' (Coco et al., 2020).

As an experiment, the course introduced a new approach framed as a material driven design approach. Instead of teaching the students to select materials for a product idea, they were given one material related to a current and relevant societal waste problem, where the task was to understand and analyse a waste material through the lens of the CE. The students were challenged to explore the material's properties through research and lab tests, to take a design approach to materials in which the idea for a product's commercial afterlife would emerge from the opportunities and limitations of the material. This method clashes with the traditional material science approach, which selects materials on a business-as-usual basis, but the course presents students with the actual environmental problems existing in the backside of the take-make-dispose model. This approach introduces students to designing for a product's commercial afterlife. Thus, students are being challenged to design using circular value chain thinking and not only focus on the product.

Normally, and in classic DT approaches, the designers select materials based on matching product requirements with material properties. In the cases described previously, it was vice versa – after selecting the materials, the application idea was based on exploring the used or recycled materials' limitations and opportunities. The Material Driven Design Method (MDDM) starts with a selected material, then designing for a commercial afterlife, or working with an

existing EOL product (e.g., wind turbine blades, upcycled textiles, or eelgrass). These were just some of the cases that we worked with.

One question to ask is: what is the product material afterlife? Sometimes, also for pedagogic reasons, to discover that the 'afterlife' options for a product are limited and often caused by some design decisions; this leaves us with the question: how could it be redesigned? As a reflection, Anastas and Zimmerman (2003) introduced, in 2003, the 12 principles of green engineering as a foundation for what is also grounded in the eco-design principles, where principle 11 is 'Designing for a commercial afterlife'. This principle focuses on designing products with the aim of recovering components that remain functional and valuable through up-front modular design, so allowing for repair, refurbishing, or sorting of materials in clean fractions for recycling. The aim of incorporating this principle in this new method is to create meaningful applications of materials with all aspects considered, especially both aspects of the material experience and aspects that ensure that the properties of the material are fully and optimally exploited throughout the product's life cycle and can be reused or recycled.

Even though both material and design industries are becoming increasingly involved in the challenge to realise the functionality and meaning of a material in a creative way, systematic approaches are currently limited when it comes to defining the experience of a material (Karana et al., 2015). Karana et al. (2015) distinguish between 'material thinking', where the technical material properties are explored as embodying a product (what it is) and material driven design, where the experience perspective is added as an extra layer to embrace the material for what it does, expresses to us, provokes in us, and makes us do.

The process of development with a MDDM is advised as an organised approach to explore and understand the material, in cases where the material is fully developed, partly developed, or entirely unknown, and thus having limited databases of the technical characteristics of the materials available. The initial experimental step of the method, engineering tinkering with the material, allows the designer to assess sensorial, interpretative, affective, and performative levels of the material through the construction of a mind-map to discover implicit motivations for using a material or composites in different applications (products). The construction of such mind-maps could also allow the designer to discover interrelationships between emotions, performances, sensorial potentials, and meanings of the material.

Material Driven Design as a method for product development is a four-staged process (see Figure 12.4). Material benchmarking is part of the experimental first step, where the material is positioned with a group of comparable materials and their corresponding applications. This is done to compare and expand the search for applicability areas. Benchmarking the material is expected not only to illuminate the potential application areas, but also lay the foundation for revealing strategies, values, or issues within the design domain, which is part of the second step. This step accentuates the current time dimension, for example, 'cradle-to-cradle' and 'design for sustainability'. After this initial step, designers have often established a probable intended use but deciding on the application of the material from this basis alone could lead to conventional solutions and utilizing only the value of the material, such as its mechanical or aesthetic properties.

How does the designer visualise the role of the material and its contribution to the performance of the potential product it embodies? Developing the material experience vision is the second step of the MDDM. This vision should address the material's purpose in relation to people, the planet, and society to recognise paths toward future applications. The mind-map, benchmarking, and vision now serve as the foundation on which to create a material experience pattern in the third step of identifying the interrelationships between the formal qualities and the experience vision of the material. Using the MDDM should be considered an iterative process that could potentially

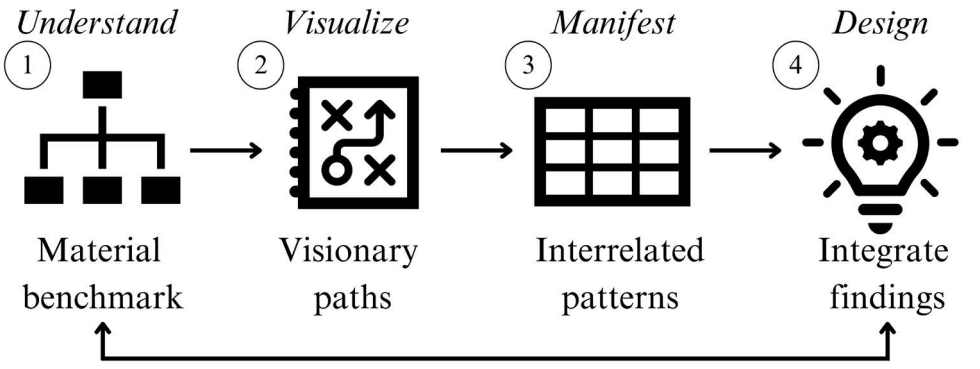


Figure 12.4 A simple illustration of the Material Driven Design Method.

require interrelated loops. In the course, we applied the use of an online materials library, a software license programme named Ansys Granta Edupack, where our students could look up each material or create a layered hybrid material and thus get information on the mechanical properties, the carbon footprint, cost, and the opportunities for reuse, recycling, downcycling, or if the created material would go to landfills or incineration. The use of this materials bank was an important step for the analysis before tinkering with the materials in practice or combining two materials.

How we did it: Teaching method

To summarise the pedagogic teaching model for combining design principles with material tinkering (the MDDM) and application in real cases, we:

- Taught the students about design principles for CE to envision diverse directions of sustainability (design lifetime, commercial afterlife, and/or recyclability) – already encountered in the decision stage.
- Invited industry case owners into the on-going process of expert knowledge sharing.
- Presented students with cases derived from real-life challenges in industry, either redesign of existing products to design for repair, technological upgradability, or new commercial afterlife of waste materials for material circularity.
- Encouraged material driven design thinking to offer a design approach grounded in material properties, and analysis of the environmental footprint and their cyclability (before initiating Design Thinking).
- Offered material test lab work and online library material, and software for obtaining knowledge about the materials, their properties, potential substitutions, and their recyclability.

Observations

A general challenge with the new teaching method was found to be the students' inherent tendency to start with creativity related to product ideas, meaning they formulated a problem using a confirmative approach: "Can the material be applied for this specific use in a commercial

afterlife?" rather than the explorative approach of sustainability that asks: "For what use can this material optimally be applied in order to secure a commercial afterlife?" It was most certainly a struggle to stay with this new method of exploring the material and let ideas grow from its properties. In the end, it was a challenge that the students reported they were proud of and learned a lot from, as we see through the implementation of these concepts in the master thesis projects that they worked on afterwards.

One of the first case challenges in the course was the recycling of wind turbine blades to avoid end disposition in a landfill. This challenge included separating chemicals and reusing composite materials like coal and fiberglass in a new, circular process. Another case challenge was to explore solid boards made from upcycled textiles using air-laid technology, which used no water. The solid boards were introduced by the company as converting textile waste materials into either acoustic felt or solid boards. Here, the students had to understand the material, its mechanical properties and its aesthetic qualities of what it could become. As well, students had to consider the materials' weaknesses, meaning what the materials could not be used for, such as the material would absorb water. An example of a common challenge reflected upon in the students' reports is the challenge of finding or gaining access to information on the products and their materials. The following example is from a case investigating a commercial afterlife for used wind turbine blades that consist of both organic materials like balsa wood, but mostly composite materials made of glass fibre enforced epoxy (plastic) in sandwiched layers, and thus are difficult to recycle:

> Transparency issues contributed to the problem because blade manufacturers keep blade data close as part of the company's intellectual property. Information on the exact material composition and the adhesives used for assembly could clarify the problem and make it easier to find possible solutions. . .

or

> Lack of knowledge around composite mechanics and calculations also proved to be restricting and time-consuming, and in combination with the uncertainty of the composites resulted in an unknown factor of the applicability of the materials. The material origin and damage potential create uncertainties of how to use these materials because planning of strength requirements is quite difficult without knowing the properties of the laminates, e.g., condition, orientation, matrix, and fibre type, etc.

This experience from teaching the course shows that design for contemporary product development reflects the systemic nature of CE with its diversity of available techniques and lack of conceptual integration to support system innovation. What is urgently required to facilitate the flow of materials at the systemic level is transparency in the supply chain and traceability in the material streams. One consideration is to introduce traceability in a circular supply chain by implementing unique identifiers in products, as also suggested by Franco (2017) or what is already done in medical device products, they have unique device identifiers (UDI) traceable to manufacturing and supply chain.

Key takeaways: The dynamics of the learning

Considering the key takeaways from theory and the reflective learning practice identified from the course's student reports, changes are required on a broad spectrum to improve the problem-based

learning approach related to designing for a CE. Clear principles of sustainability are relevant tools for the students, and a material driven design perspective is necessary for an explorative approach to what decisions we as designers make. Such decisions will have consequences for the environment and for either hindering or creating possibilities for product or material recirculation. We need new approaches that can relate to the challenges of circularity from today's product design – and which include both EOL challenges, material flows, and solutions that integrate the environmental impact with the social and economic impact.

First, we experience that the students gained an understanding of the meaning of sustainable design and the design principles this involves. Second, and more importantly, the students also gained an understanding of how important their role as designers is for sustainable products. Third, through the business-driven cases used in the course, the students also were able to implement the knowledge they gained in the course to solve real sustainability challenges.

Every year, we take in new cases, as this provides not only flexibility but also enables us to develop the course and the methodology to moving from a problem-centred method to a flexible problem-solution process that recognises the explicit duality of human rationality and behaviour. Because as we see it, solutions must reflect the dynamics of the times we live in and challenge the assumptions.

Perspectives on design as a catalyst for socio-technical system transformation

Clearly, the role of design in the transition towards CE does not begin and end at the level of product or service innovation; rather, it restarts with the synthesis of the economic transition. As Banathy (2013, pp. 1–2) suggested already in 2013, "In an age when speed, intensity, and complexity of change increases constantly and exponentially, the ability to shape change, rather than being its victims or spectators, depends on our competence and willingness to guide the purposeful evolution of our systems, our communities, and our society." This notion leads us to bring design into a discussion of socio-technical systems.

The fundamental transformation of a socio-technical system is the result of multiple processes. It is a transition unfolding over several decades, leading to extensive changes across different technological, material, organizational, institutional, political, economic, and socio-cultural dimensions (Markard et al., 2012). During a socio-technical transition, new functions, processes, products, services, business models, and organizations emerge to complement or replace existing ones. At this level, design approaches focus on promoting radical changes on how societal needs are fulfilled through technological, social, organizational, and institutional innovations (Ceschin & Gaziulusoy, 2016). However, it is important to note that in practice a socio-technical transition, for example, the practice of moving towards sustainable development, is a wicked problem (Buchanan, 1992), and there is little experience designing transitions at such a level of complexity.

The process of sustainable design is an evolutionary one, but it can be stimulated by changes at the meso-level created when new problems arise, for example, public resistance to pollution or price increases for scarce resources. There can be changes in technology or changes in the landscape pushed by policy; we have argued that the EU Ecodesign Directive or the EU Lead-free Directive should apply for all product categories and not select exceptions. Transformations occur when developments at multiple levels strengthen each other in a new form (Geels & Kemp, 2006), resulting in dynamic transitions between industries, new policies or regulations, new networks, and value chains, or even because of industrial symbioses.

A particularly difficult, yet critical task when designing transitions is the need to anticipate and ideate the conditions of the system we want to achieve; an ideal system that does not yet exist. Transitioning from present to future states can be either the result of an unintended effect or a rational endeavour. If an intended effect, we need to design a roadmap for actions to achieve the desired future; a strategic set of plans defining roles, goals, and timelines needs to be generated. Although design inherently always looks into the future by producing something that does not yet exist, we argue that the abstraction required to define alternative futures is not an explicit function of the design process, but rather an implicit trait of design thinking. Gordon et al. (2019) suggest that, as a problem-solving approach, DT inherently looks into the future of potential solutions. We argue that education-informed design has ample application in system innovation, notably for the ideation and synthesis of alternative distant future socio-technical scenarios and potential roadmaps for transition to a CE; in this context, inclusive participation and collaboration of actors and stakeholders is in the process of co-creation of alternative futures. Wilkinson et al. (2014, p. 1) conclude that large-scale changes in socio-technical and political conditions can be induced through "open and collaborative innovation processes in which diverse interests collaborate to co-create the future."

Design processes can be both participatory and exploratory, and they can internalise the knowledge, interests, and expectations of multiple actors within the socio-technical system. By enabling the co-creation of alternative narratives to assess past and present conditions and to devise future outcomes, design can act as a democratic platform for social learning, a crucial requirement for effective system innovation (Wilkinson et al., 2014). In retrospect, we can conclude that the relationship between design and change has a dual nature: design changes the artificial and the natural, but also reflects such changes in the constant evolution of its meaning and use. The theory and practice of design, therefore, both internalises and externalises the changes in our collective evolution.

Design can change the fast consumer product trends, an almost neophiliac culture, when it shifts from designing products for short life cycles to longer life cycles. However, as a process to guide the required innovation of systems, the current design methodology is not sufficient. It must be supplemented with the theory and practice of other disciplines as we seek long-term sustainable solutions. We argue that a systemic innovation view of design is needed; it couples design thinking with future planning to support the vision of decoupling economic growth from the increasing resource use and boost sustainable growth. One way to achieve this is to introduce foresight methodology that focuses on participatory processes and techniques, such as exploring EOL scenarios and co-created roadmaps for developing solutions across the entire value chain, from design to manufacturing to full circularity. We see these steps as more open innovation approaches with interesting possibilities for both the theory and practice of system innovation. The how-to includes an emphasis on coupling the design process and the strategic policy process at both the macro- and micro-levels as well as strategic decision-making processes in business organizations. A catalyst from a design perspective for a CE, in this sense, is not just one thing. Multiple drivers and forces with an impact on product design choices can become a catalyst if they are aligned with a vision of a sustainable future based on dynamic learning processes.

Conclusions: The full circle

Product design is a highly adaptive process at the micro-level of system innovation, responding dynamically to new conditions at the meso- and macro-levels. But we also need to focus on the inner transition and reform our education by teaching design that integrates sustainability and

circularity to show how we can get to a sustainable outcome. We will then be able to fully utilise design as a catalyst for CEs in a sustainable future.

We conclude that the role of design as a catalyst of change, as a rational and creative process, can guide the development of valuable products and services to design for circularity. We have argued that, in the context of a CE, the strategic value of design as a catalyst of change lies in creating and accelerating movement towards a sustainable future. This requires the co-creation of a future CE through collaborative and inclusive definition of distant scenarios for sustainable systems of production and consumption, which can elicit new narratives to discuss and define the new directions and timelines our current socio-technical systems could follow. We see a need to produce clear visions of what the future of a sustainable CE should be, including action plans for aligning policy and product standards with the application of eco-design principles.

As a catalyst of change, design overcomes inertia and creates movement, operating simultaneously at all levels of socio-technical systems as a quality, function, process, and product. Thus, design can modify behaviour, guide strategies and decisions, set and reflect patterns and trends, fulfil and create needs, visualise and materialise ideas, solve problems, innovate functions and processes, and transform the material and immaterial configuration of the human-made world.

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